

Two Gaussian counterexamples to proposed sharp constants for Heisenberg logarithmic Sobolev inequalities

Counterexample packet for arXiv:2410.15566

Agent `agent_lane_02` (GPT5.6)

9 August 2026

Abstract

We give two explicit normalized Gaussian counterexamples. First, on $\mathbb{H}^2$ in the convention of Antonelli–Calzi–Gordina, a Gaussian violates Corollary 2.3 when that corollary is instantiated with the constant printed in Remark 2.2. The cause is a transcription error: the cited sharp Folland–Stein theorem has gamma-ratio exponent $2/Q$, whereas the source prints $1/Q$. Second, on $\mathbb{H}^1$ in Suguro’s convention, a Gaussian proves that $2/(Q\pi e)$ cannot equal the optimal logarithmic Sobolev constant. Both comparisons are exact and elementary. These counterexamples do not determine the optimal constant asked for in Question 2.4, which remains open, and the second does not address Suguro’s separate asymptotic suggestion.

1 Claims being tested

Antonelli–Calzi–Gordina use an H-type group with first-layer dimension $2n$, center dimension m , homogeneous dimension $Q = 2n + 2m$, and Lebesgue Haar measure. Their Remark 2.2 prints the constant

$$C_{n,m}^{\text{print}} = \frac{4^{2m/Q}}{2n(Q-2)\pi^{(2n+m)/Q}} \left(\frac{\Gamma(2n+m)}{\Gamma((2n+m)/2)} \right)^{1/Q}. \quad (1)$$

Their Corollary 2.3 then states, for $\int f^2 = 1$,

$$\text{Ent}(f^2) \leq \frac{Q}{2} \log \left(C_{n,m} \int |\nabla_H f|^2 \right). \quad (2)$$

Figure 1 reproduces the relevant source text.

For the Heisenberg group ($m = 1$), the same paper asks for the best constant α_n in this inequality (Figure 2). A second paper, Suguro (2024), proposes that in its standard Heisenberg normalization the best constant either equals $2/(Q\pi e)$ or approaches it in high dimension; see Figure 3. We disprove the exact equality already in dimension three. We make no claim about the asymptotic branch.

2 Proof intuition

For any normalized test function, a logarithmic Sobolev inequality of the form

$$\text{Ent}(f^2) \leq \frac{Q}{2} \log(C\mathcal{E}(f))$$

forces

$$C \geq J(f) := \frac{\exp(2 \text{Ent}(f^2)/Q)}{\mathcal{E}(f)}.$$

$(x_1, \dots, x_n, y_1, \dots, y_n, z)$. A basis of left-invariant vector fields is given by

$$X_i = \partial_{x_i} + \frac{y_i}{2} \partial_z, \quad Y_i = \partial_{y_i} - \frac{x_i}{2} \partial_z, \quad \text{for any } i = 1, \dots, n, \quad (2.1)$$

$$Z = \partial_z. \quad (2.2)$$

2.2. Sobolev inequalities on H -type groups. We will use the following result which has been very recently proved in [35, Theorem 1.2]. It is a generalization to H -type groups of a celebrated result by Jerison–Lee proved for $\mathbb{H}^n$ in [24].

Theorem 2.1 ([24, 35]). *For every $m, n \geq 1$, there exists a constant $C_{n,m}$ such that the following holds. For every H -type group $\mathbb{G}$ with $\dim(\mathfrak{v}) = 2n$, and $\dim(\mathfrak{z}) = m$, and for every $f \in W^{1,2}(\mathbb{G}, d\mu)$ we have*

$$\left(\int_{\mathbb{G}} |f|^{\frac{2Q}{Q-2}} d\mu \right)^{\frac{Q-2}{Q}} \leq C_{n,m} \int_{\mathbb{G}} |\nabla f|^2 d\mu, \quad (2.3)$$

where $Q := 2n + 2m$.

Remark 2.2. The value of the sharp constant $C_{n,m}$ and the expression of the minimizers for arbitrary H -type groups are computed explicitly in [35, Theorem 1.2]. Namely,

$$C_{n,m} := \frac{4^{2m/Q}}{2n(Q-2)\pi^{(2n+m)/Q}} \left(\frac{\Gamma(2n+m)}{\Gamma(\frac{2n+m}{2})} \right)^{1/Q}.$$

As a corollary of the previous estimate, following Beckner–Pearson [7], we can deduce the following. Compare with the recent [12, Corollary 3.6]. For discussions on the optimality of the constant $C_{n,m}$ in Corollary 2.3, see Question 2.4.

Corollary 2.3. *Let $\mathbb{G}$ be an H -type group. Let $C_{n,m}$ be the constant in (2.3) in Theorem 2.1. For every $f \in W^{1,2}(\mathbb{G}, d\mu)$ with $\int_{\mathbb{G}} f^2 d\mu = 1$ we have*

$$\int_{\mathbb{G}} f^2 \log(f^2) d\mu \leq (n+m) \log \left(C_{n,m} \int_{\mathbb{G}} |\nabla f|^2 d\mu \right). \quad (2.4)$$

Moreover, if there exist $C \in \mathbb{R}$ and $D > 0$ such that for every $f \in W^{1,2}(\mathbb{G}, d\mu)$ with $\int_{\mathbb{G}} f^2 d\mu = 1$ we have

$$\int_{\mathbb{G}} f^2 \log(f^2) d\mu \leq C \log \left(D \int_{\mathbb{G}} |\nabla f|^2 d\mu \right), \quad (2.5)$$

Figure 1: Remark 2.2 and Corollary 2.3 of arXiv:2410.15566v1. The gamma-ratio exponent printed in the boxed source constant is $1/Q$.

Thus one explicit function with $J(f) > C$ is a complete counterexample. Product Gaussians are especially convenient: normalization and entropy are immediate, while the cross terms introduced by the two horizontal vector fields cancel. A suitable central variance then makes each desired strict comparison transparent.

3 The two counterexamples

Theorem 1 (Two explicit Gaussian counterexamples). *The following statements hold.*

- (i) *The constant $C_{n,m}^{\text{print}}$ in (1) does not make (2) true. More precisely, the case $n = 2$, $m = 1$ is violated by the normalized Gaussian in (3).*
- (ii) *In Suguro’s standard convention on $\mathbb{H}^1$, the optimal constant in the scale-invariant logarithmic Sobolev inequality is strictly larger than $2/(Q\pi e) = 1/(2\pi e)$. The normalized Gaussian in (9) is an explicit witness.*

and finally observe that $p/(p-z) = n+m$ since $Q = zn + zm$.

The fact that $(n+m)$ is the only possible C in (2.5) comes from a scaling argument. Calling $f_\lambda(x) := \lambda^{Q/2} f(\delta_\lambda(x))$ we get that, for every $\lambda > 0$, $\int_{\mathbb{G}} f^2 d\mu = 1$ if and only if $\int_{\mathbb{G}} f_\lambda^2 d\mu = 1$. Moreover, for every f such that $\int_{\mathbb{G}} f^2 d\mu = 1$, we have

$$\int_{\mathbb{G}} f_\lambda^2 \log(f_\lambda^2) d\mu = Q \log \lambda + \int_{\mathbb{G}} f^2 \log(f^2) d\mu,$$

and

$$\int_{\mathbb{G}} |\nabla f_\lambda|^2 d\mu = \lambda^2 \int_{\mathbb{G}} |\nabla f|^2 d\mu.$$

Thus, the validity of (2.5) for every f_λ with $\lambda > 0$ forces $2C - Q = 0$, from which the assertion follows. $\square$

We stress here the following question, which, taking into account Lemma 3.1, appears to be related to finding the best constant in (1.1).

Question 2.4. *Find the best (smallest) constant α_n for which the following holds. For every $f \in W^{1,2}(\mathbb{H}^n, d\mu)$ with $\int_{\mathbb{H}^n} f^2 d\mu = 1$ we have*

$$\int_{\mathbb{H}^n} f^2 \log(f^2) d\mu \leq (n+1) \log \left(\alpha_n \int_{\mathbb{H}^n} |\nabla f|^2 d\mu \right). \quad (2.7)$$

Remark 2.5. For the Euclidean version of Question 2.4 see [13], and the recent [5]. In the Euclidean case, (2.7) is equivalent to Gross' logarithmic inequality, so that optimality is achieved by replacing $n+1$ with $n/2$, the sharp constant being $\alpha_n := 2/(\pi n e)$.

In the Heisenberg group a version of (2.7) has been obtained in [19, Corollary 2.5] as a consequence of differentiating a sharp Hardy–Littlewood–Sobolev inequality on $\mathbb{H}^n$. Anyway, to the best of our knowledge, it is not known whether [19, Corollary 2.5] gives information

Figure 2: Question 2.4 and Remark 2.5 of arXiv:2410.15566v1.

3.1 Counterexample to the printed source constant

Use the source coordinates on $\mathbb{H}^2 = \mathbb{R}_x^2 \times \mathbb{R}_y^2 \times \mathbb{R}_z$:

$$X_i = \partial_{x_i} + \frac{y_i}{2} \partial_z, \quad Y_i = \partial_{y_i} - \frac{x_i}{2} \partial_z, \quad Q = 6.$$

Let $r^2 = |x|^2 + |y|^2$ and set

$$f(x, y, z) = \pi^{-1} \left(\frac{8}{5\pi} \right)^{1/4} \exp \left(-\frac{r^2}{2} - \frac{4}{5} z^2 \right). \quad (3)$$

Its square is a product probability density, so $\int f^2 = 1$. Direct Gaussian integration gives

$$\text{Ent}(f^2) = -\frac{5}{2} \log \pi + \frac{1}{2} \log \frac{8}{5} - \frac{5}{2}, \quad \mathcal{E}(f) = \int \sum_{i=1}^2 ((X_i f)^2 + (Y_i f)^2) = \frac{12}{5}. \quad (4)$$

For completeness, if

$$f_{a,b} = \left(\frac{a}{\pi} \right)^{n/2} \left(\frac{b}{\pi} \right)^{1/4} e^{-ar^2/2 - bz^2/2},$$

then

$$X_i f_{a,b} = \left(-ax_i - \frac{b}{2} y_i z \right) f_{a,b}, \quad Y_i f_{a,b} = \left(-ay_i + \frac{b}{2} x_i z \right) f_{a,b}.$$

The mixed terms cancel, and the Gaussian moments yield

$$\text{Ent}(f_{a,b}^2) = n \log \frac{a}{\pi} + \frac{1}{2} \log \frac{b}{\pi} - n - \frac{1}{2}, \quad \mathcal{E}(f_{a,b}) = na + \frac{nb}{8a}.$$

which implies the desired inequality (1.5). $\square$

Remark In Euclidean space $\mathbb{R}^n$, by the Stirling approximation for the Gamma function

$$\Gamma(z) \approx (2\pi)^{\frac{1}{2}} e^{-z} z^{z-\frac{1}{2}} \quad \text{for } z \gg 1,$$

the sharp constant S_n in (1.1) is, for n large enough:

$$\begin{aligned} \frac{n\pi e}{2} S_n &= \frac{1}{n-2} 2^{1-\frac{2}{n}} \pi^{-\frac{1}{n}} e \Gamma\left(\frac{n+1}{2}\right)^{\frac{2}{n}} \\ &\approx \frac{1}{n-2} 2^{1-\frac{2}{n}} \pi^{-\frac{1}{n}} e \left(\frac{4\pi}{n+1}\right)^{\frac{1}{n}} \left(\frac{n+1}{2e}\right)^{1+\frac{1}{n}} \quad \text{for } n \gg 1 \\ &= \frac{n+1}{n-2} (2e)^{-\frac{1}{n}} \approx 1 \quad \text{for } n \gg 1, \end{aligned}$$

which implies that the sharp constant S_n in (1.1) is approximated by the one in the logarithmic Sobolev inequality $2/(n\pi e)$ for n large enough.

On the other hand, in the Heisenberg group $\mathbb{H}^n$, the constant $S_{\mathbb{H}}$ in (1.4) is approximated by $2/(Q\pi e)$ as the following:

$$\begin{aligned} \frac{Q\pi e}{2} S_{\mathbb{H}} &= \frac{Q}{(Q-2)^2} 2^{1-\frac{4}{Q}} e \Gamma\left(\frac{Q}{2}\right)^{\frac{2}{Q}} \\ &\approx \frac{Q}{(Q-2)^2} 2^{1-\frac{4}{Q}} e \left(\frac{4\pi}{Q}\right)^{\frac{1}{Q}} \frac{Q}{2e} \quad \text{for } Q \gg 1 \\ &= \frac{Q^2}{(Q-2)^2} \left(\frac{\pi}{4Q}\right)^{\frac{1}{Q}} \approx 1 \quad \text{for } Q \gg 1 \end{aligned}$$

when n is large enough. By the above observation, we conjugate that the sharp constant in (1.5) coincides with $2/(Q\pi e)$ or approximates it for n large enough.

As a corollary, one can obtain the parametric logarithmic Sobolev inequality on the Heisenberg group. The following inequality is useful for estimating solutions to the heat equation on the Heisenberg group:

Figure 3: The concluding proposal following Theorem 1.2 in Suguro (2024): exact coincidence with $2/(Q\pi e)$ or asymptotic approximation.

Equations (4) are the specialization $n = 2$, $a = 1$, $b = 8/5$.

The necessary constant forced by this single test function is

$$J(f) = \frac{\exp(\text{Ent}(f^2)/3)}{\mathcal{E}(f)} = \frac{5}{12} \left(\frac{8}{5}\right)^{1/6} (\pi e)^{-5/6} = 0.0754405987422 \dots \quad (5)$$

On the other hand, substituting $n = 2$, $m = 1$, $Q = 6$ in the source's printed formula gives

$$C_{2,1}^{\text{print}} = \frac{4^{1/3}}{16\pi^{5/6}} \left(\frac{\Gamma(5)}{\Gamma(5/2)}\right)^{1/6} = 2^{-5/2} \pi^{-11/12} = 0.0619019448788 \dots \quad (6)$$

The comparison is exact. Indeed,

$$\frac{J(f)}{C_{2,1}^{\text{print}}} = \frac{10}{3} 5^{-1/6} \pi^{1/12} e^{-5/6} > 1. \quad (7)$$

To see the last inequality without decimals, note that $5^{-1/6} > 3/4$ because $(4/3)^6 > 5$, that $\pi^{1/12} > 1$, and that $e^{-5/6} > 2/5$. The last bound follows, for example, from $e < 11/4$ and $(11/4)^5 < (5/2)^6$. The

product of the three displayed lower bounds with $10/3$ is 1, and all inequalities are strict. Therefore $J(f) > C_{2,1}^{\text{print}}$, equivalently

$$\text{Ent}(f^2) > 3 \log \left(C_{2,1}^{\text{print}} \mathcal{E}(f) \right),$$

which proves Theorem 1(i).

3.2 Why the source constant is too small

Yang's cited Theorem 1.2 has exponent $2/Q$, not $1/Q$, on the gamma ratio. In the source notation the correctly transcribed sharp Folland–Stein constant is

$$\tilde{C}_{n,m} = \frac{4^{2m/Q}}{2n(Q-2)\pi^{(2n+m)/Q}} \left(\frac{\Gamma(2n+m)}{\Gamma((2n+m)/2)} \right)^{2/Q}. \quad (8)$$

Thus the failure above is not a gap in the Jensen argument of Corollary 2.3; it is a numerical transcription error in the input Sobolev constant. For $m = 1$, the duplication formula reduces (8) to

$$\tilde{C}_{n,1} = \frac{(n!)^{1/(n+1)}}{\pi n^2}.$$

The counterexample does not violate the corollary after this correction.

3.3 Counterexample to exact equality with $2/(Q\pi e)$

Now use Suguro's convention on $\mathbb{H}^1 = \mathbb{R}^3$:

$$X = \partial_x + 2y\partial_s, \quad Y = \partial_y - 2x\partial_s, \quad Q = 4,$$

with Haar measure $dx dy ds$. Define

$$g(x, y, s) = \pi^{-1/2} \left(\frac{1}{6\pi} \right)^{1/4} \exp \left(-\frac{x^2 + y^2}{2} - \frac{s^2}{12} \right). \quad (9)$$

Again $\int g^2 = 1$. Direct calculation gives

$$\text{Ent}(g^2) = -\frac{3}{2} \log \pi - \frac{1}{2} \log 6 - \frac{3}{2}, \quad \mathcal{E}(g) = \int ((Xg)^2 + (Yg)^2) = \frac{4}{3}. \quad (10)$$

Indeed, for the general Gaussian with exponents $-a(x^2 + y^2)/2 - cs^2/2$, the same cancellation gives energy $a + 2c/a$; here $a = 1$ and $c = 1/6$.

Consequently this single Gaussian forces

$$J(g) = \frac{\exp(\text{Ent}(g^2)/2)}{\mathcal{E}(g)} = \frac{3}{4} (6\pi^3 e^3)^{-1/4} = 0.0959268502867 \dots \quad (11)$$

Suguro's proposed exact value is

$$\frac{2}{Q\pi e} = \frac{1}{2\pi e} = 0.0585498315243 \dots$$

Their exact ratio is

$$\frac{J(g)}{1/(2\pi e)} = \frac{3}{2} \left(\frac{\pi e}{6} \right)^{1/4} > 1, \quad (12)$$

because $\pi > 3$ and $e > 2$ imply $\pi e > 6$. Hence a logarithmic Sobolev inequality with constant $1/(2\pi e)$ fails for the explicit g in (9), proving Theorem 1(ii).

4 Additional partial information and limitations

Optimizing the source-convention family $f_{a,b}$ over its genuine anisotropy b/a^2 gives the lower bound

$$\alpha_n \geq \frac{2n+1}{2n(n+1)} \left(\frac{8}{2n+1} \right)^{1/(2(n+1))} (\pi e)^{-(2n+1)/(2(n+1))}.$$

Together with the corrected Jensen–Folland–Stein upper bound, this yields

$$\frac{2n+1}{2n(n+1)} \left(\frac{8}{2n+1} \right)^{1/(2(n+1))} (\pi e)^{-(2n+1)/(2(n+1))} \leq \alpha_n \leq \frac{(n!)^{1/(n+1)}}{\pi n^2}.$$

This bracket is useful but does not determine α_n . The second counterexample refutes only exact coincidence with $2/(Q\pi e)$; it does not rule out asymptotic approximation as $n \rightarrow \infty$.

5 Verification and bounded novelty search

The accompanying script independently evaluates both counterexample ratios, checks the corrected gamma formula, and reproduces the Gaussian-family comparison in dimensions 1 through 20. The source statements were transcribed from the locally archived PDFs, and the three evidence crops above retain the complete width of the relevant pages.

The run’s cheap indexes were searched by arXiv id, title, and the phrases “ $2/(Q\pi e)$ ”, “gamma ratio”, “erratum”, “correction”, and “Gaussian counterexample”. A bounded primary-source web search through 9 August 2026 checked the current arXiv record for Antonelli–Calzi–Gordina, Yang’s cited arXiv theorem, and Suguro’s journal article. No erratum, prior correction, or prior Gaussian disproof was found. This supports candidate novelty only; it is not an exhaustive literature claim.

6 Human-review recommendation

Both counterexamples are short, explicit, and likely valid. Before external circulation, a human should independently check (a) that (1) is the exact constant intended in Corollary 2.3, (b) the notation map from Yang’s dimensions to $(2n, m)$, and (c) that Suguro’s proposal is read as the disjunction reproduced in Figure 3. The scope distinction between disproving the exact-equality branch and leaving the asymptotic branch open should be preserved.

References

1. G. Antonelli, M. Calzi, and M. Gordina, *Sharp defective log-Sobolev inequalities on H-type groups*, arXiv:2410.15566v1, Remark 2.2, Corollary 2.3, and Question 2.4.
2. Q. Yang, *The optimal constant in the L^2 Folland–Stein inequality on the H-type group*, J. Funct. Anal. 286 (2024), Paper 110209; arXiv:2301.03332v4, Theorem 1.2.
3. T. Suguro, *The logarithmic Sobolev inequality on the Heisenberg group and applications to the uncertainty inequality and heat equation*, J. Inequal. Appl. 2024, Paper 99, Theorem 1.2 and the following remark.